

Comorbidities

In male patients, **lichen sclerosus (LS)** is associated with an increased body mass index and has been linked to **coronary artery disease**, **diabetes mellitus (DM)**, and **tobacco use**.

Diabetes mellitus and LS co-occur more frequently than expected by chance, leading some experts to recommend **screening for DM in patients with LS**.

Women with LS often report additional bladder, bowel, and pain-related conditions. In a cohort of 308 women with LS attending a vulval clinic, self-reported comorbidities included: - **Overactive bladder** (15.3%) - **Stress urinary incontinence** (27.9%) - **Constipation** (32.5%) - **Irritable bowel syndrome** (19.5%) - **Thyroid dysfunction** (33.1%) — representing a 2- to 3-fold increased risk - **Fibromyalgia** (9.1%) - **Temporomandibular joint disorder** (13.0%) - **Vulval pain** (83.1%)

A multicentre Italian study suggests that **metabolic factors**—including obesity, hypertension, hypothyroidism, and sedentary lifestyle—may contribute to the development of genital LS in genetically predisposed individuals, with a similar risk profile observed in both males and females.

Hu et al. compared women with vulval LS (n = 865) to those with other vulval conditions (n = 1118). **Increasing age**, **thyroid disease**, and **anorectal fissures** were significantly associated with vulval LS. The link between anorectal fissures and LS is likely a **sequela of the disease** rather than a true comorbidity.

An association between **genital melanoma** and LS has been reported in multiple case series, particularly in women. **Vulval melanoma** is rare, with an incidence of 0.10–0.13 per 100,000 individuals, typically affecting postmenopausal women. However, patients with LS appear to have an **increased risk** of developing vulval melanoma, including in rare cases among

girls. Additionally, there is an elevated risk of **squamous cell carcinoma (SCC)** in LS patients, which should be carefully monitored.

EPIDEMIOLOGY

Incidence/Prevalence

The exact prevalence of LS remains uncertain and is likely **underestimated**, partly due to underdiagnosis. It is a common condition seen in vulval and penile dermatology clinics. Alongside spongiotic dermatitis and other lichenoid interface dermatoses, LS accounts for the majority of foreskin specimens examined in pathology departments.

Estimated prevalence ranges from **0.1% in children** to **3% in women over 80 years**. Data from the Oxford clinic suggest an annual presentation rate of approximately **150–200 women per million population**. A recent study found that by age 80, **1.6% of women** will have developed LS. In males, the estimated incidence is **0.07%**, based on retrospective review of discharge records from a US Army Medical Center.

Age at Onset

LS can develop at any age. The prevalence of **vulval LS increases with age**, with the highest frequency observed in **postmenopausal women**, although this may not be directly tied to menopausal status alone. Approximately **20% of affected women experience onset before menopause**.

A **second peak** in incidence is observed **before puberty**, with both peaks occurring during periods of **low oestrogen levels**. This has led to speculation that low oestrogen may play an aetiological role, possibly through modulation of immune responses—oestrogen tends to favor T-cell-mediated immunity over antibody-mediated responses. Alternatively, reduced lubrication during these

periods may increase susceptibility to **mechanical trauma** (Koebner phenomenon).

Wallace described a **continuous rise in incidence from puberty to perimenopause**, peaking around menopause, followed by a decline. In men, incidence rises after puberty, possibly with a **prepubertal peak**, and decreases again after age 60. One study reported a near doubling of incidence in the third decade, which may reflect the military setting of the study, though a similar peak has also been observed in non-military populations.